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Blessing or curse? Appreciation, amenities and resistance to urban renewal

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ABSTRACT

This article investigates the 2008 referendum held in opposition against the "Mediaspree", a major urban development project in Berlin that has been perceived as a threat of displacement of local residents and culture. Using precinct level data we find a high degree of localized resistance around the project area, conditional on socio-demographic characteristics. A comparison to local appreciation rates shows that in an environment of very low owner occupancy public (re)development projects are opposed the more residents associate them to an increase in area valuation. This effect is, however, not strong enough to explain the localized resistance. Considering a micro-level data set on music nodes, our results suggest that this effect is instead attributable to a feared loss of specific cultural amenities and neighborhood character.

1. Introduction

In recent decades residential downtown areas in the U.S. as well as Europe have experienced a remarkable comeback after a long period of steady decline accompanied by the migration of middle- and upper-class residents to the city fringe. The rediscovery of downtown areas has been explained in economic terms by a relative reduction in transport costs for the poor, which has reduced the comparable advantage of the rich at the city fringe (LeRoy and Sonstelie, 1983), or by the modernization of downtown housing stock, which increases the demand of high-income households (Brueckner and Rosenthal, 2005). In addition, lifestyle pluralization (Veal, 1993) has produced new upper and middle social classes such as Florida (2002) prominent *creative class*, whose highly skilled members appreciate the vitality as well as the cultural and social diversity of the inner-cities. At the same time, ambitious authorities have developed revitalization strategies in order to promote the so-called *urban renaissance* (DETR, 1999).

The (re-)invasion of residential downtown areas by high and middle classes – usually called *gentrification* – has raised many questions about what happens to the established residents. Within an environment of high owner occupancy, which is typical for most of the U.S., homevoters generally tend to support public projects that are expected to raise property values (Brunner and Sonstelie, 2003; Dehring et al., 2008; Fischel, 2001). Downtown redevelopment strategies, however, often induce fear and anger among the resident population, particularly in rental markets. On the one hand, residents may experience a demand driven increase in the cost of living space and, hence, displacement pressures. On the other hand, the inflow of higher income residents and the appreciation of the neighborhood may lead to an adjustment in local services and entertainment opportunities and loss of cultural networks, which have constituted a specific neighborhood character. In other words, residents are driven out of their consumption optima as they are forced to consume not only too much, but also the wrong neighborhood quality. A detailed understanding of the origins of such tensions in revitalization areas is crucial to enable authorities to better target the needs of local residents in *urban renaissance* strategies. Quantitative analyses, however, are complicated by the requirement of detailed data that capture perturbation among residents as well as their socio-economic characteristics, local appreciation and specific amenities that constitute the neighborhood character and are typically difficult to find.

These preconditions are met for the study area of the subject case, the "Mediaspree", a riverside urban development project within the eastern downtown area in Berlin, Germany. To both riversides of the Spree, the project is embedded into adjoining areas where the major German urban redevelopment programs "Stadtumbau Ost" and "Stadtumbau West" operate, which share the basic ideas with many international counterparts, e.g. in the U.S. or the U.K. (DETR, 1999; HUD, 1999). The project is located within the district Friedrichshain–Kreuzberg, which has become one of the avant-garde areas in post-unification Berlin. The neighborhood alongside the riversides, which offered plenty of unoccupied space, has emerged as an internationally recognized cluster of alternative music culture. The mixture of large-scale urban development programs encompassing considerable public investment and a fertile cultural environment eventually attracted international music enterprises like Universal and MTV as well as households belonging to higher income classes, the so-called *gentrifiers*. While gentrification, compared to international examples, e.g. lower Manhattan, was still at an early stage,

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the Mediaspree project by the end of the 2000s was increasingly perceived as an accelerator of neighborhood change and a threat to neighborhood stability. From this relatively typical conflict a strong neighborhood resistance movement emerged. A citizens' initiative eventually enforced a public referendum against the Mediaspree, which was held in 2008 and was supported by as much as 87% of the voters who engaged in the referendum. A further interesting feature of our study area is an extremely low rate of owner occupancy, which increases the typical conflicts between the traditional lower-class inhabitants, pioneers and gentrifiers.

In this paper we take the opportunity to investigate the main driving forces of neighborhood resistance on the basis of the stated preferences of tens of thousands of voters participating in the referendum. We conduct a spatial precinct level analysis that, besides socio-demographic characteristics and political orientation of the resident population, considers the local evolution of apartment prices and the local endowment with specific cultural amenities. We do this to evaluate whether a) there is evidence of a local increase in demand for living space induced by the development and revitalization projects, and b) whether local appreciation as an objective measure of displacement pressure explains the spatial pattern of resistance revealed by the referendum or c) if resistance is driven by an expected disutility from a loss of neighborhood character. We approximate the specific neighborhood character using a new data set on music nodes, which are used for the first time in a spatial statistical analysis.

The next section provides a more detailed description of our study area on the background of a simple gentrification model. Our empirical analysis follows in a tripartite structure, focusing on spatial patterns in the voting results (Section 3.1), appreciation rates (Section 3.2) and the interaction of both as well as the role of the local cultural geography (Section 3.3).

2. The study area

Over recent decades, a huge body of theoretical and empirical gentrification literature has emerged. Although different academic disciplines, town planners, or journalists, have developed their particular perspectives on the phenomenon, displacement of local residents and a change of a neighborhood's character due to an inflow of higher income/status households clearly emerge as a common theme from the literature (e.g. Atkinson, 2004; Freeman and Braconi, 2004; Marcuse, 2005). Accordingly, the inflow of wealthier residents brings with it increasing housing prices and rent-levels, which mirror the increased attractiveness of the area and puts pressure on households facing tighter budget constraints.[1] Over the decades the literature on gentrification has become more and more extensive and sophisticated, addressing both demand (e.g. the *emancipatory city*) and supply (e.g. the *revanchist city)* side explanations (Butler, 1997; Caufield, 1994; Ley, 1996; Smith, 1996) as well as an increasing number of more specific issues like super-gentrification, ethnic minority gentrification or the effect of gentrification on participation in democracy (Knotts and Haspel, 2006).[2] Urban economics research searches for determinants of gentrification on the basis of formal models, paying special attention to the age and quality of housing stock (Brueckner and Rosenthal, 2005) and the affordability of urban high-speed transport (LeRoy and Sonstelie, 1983).

It is clearly beyond the scope of this contribution to provide a comprehensive survey on gentrification research. For the purposes of this study it should be sufficient to describe the situation within our study area on the background of the fairly simplistic stage models, which more or less build on the work of Clay (1979) and still feature prominently in the German gentrification literature. Accordingly, there is a pioneering generation of risk affine singles or two-person households without children, typically students or creative professionals, who have a relatively low-income but a higher education compared to the residents living within the neighborhood to which they immigrate. Usually, pioneers feel attracted by relatively affordable living space within highly accessible inner-city neighborhoods with an urban fabric of pre-World War II buildings, which will typically be in bad physical condition. These minimum criteria are easily met by our subject area, which is connected by numerous underground, suburban railway and streetcar lines and whose building stock is largely formed by downtown blocks developed at the end of the 19th century. The riverbanks of the Spree, which represents the natural border between Friedrichshain and Kreuzberg as well as former East- and West-Berlin, furthermore represent a highly attractive natural amenity. While the pioneer stage of gentrification in Friedrichshain began no earlier than after the fall of the Iron Curtain, it is notable that Kreuzberg, by the time of Germany's unification, was not only inhabited by classical working-class and migrant milieus, but also represented a center of left-side politicized milieus with roots in the 1970s' autonomists scene. Initially, the neighborhood change after unification occurred at a relatively low speed, in particular on the Kreuzberg side, since the downtown areas of Mitte and Prenzlauer Berg was the focal point of pioneers who moved to Berlin in search of a vivid and cutting-edge lifestyle. As these areas entered the second stage of gentrification and increasingly attracted higher income gentrifiers, the pioneers turned their attention to Friedrichshain and Kreuzberg. The riverbanks offered large attractive areas that, particularly on the Friedrichshain side, were only sparsely developed and qualified as very attractive sites for alternative cultural activities with an initially temporal character.

To understand the character of the neighborhood it is important to note that, at the same time, Berlin was experiencing a social elevation as one of the most important hotspots for – in particular electronic – music and club culture, which increasingly occupied the open spaces and abandoned industrial buildings along the riversides within our study area. By the mid-2000s the border area between the districts had finally developed to one of the densest clusters of contemporary electronic music in the world. Besides music clubs such as Berghain and the Watergate, which both feature among the top ranked clubs world-wide (DJMAG, 2009), clubs like the Maria, which has long collaborated with the international arts and media festival *transmediale*, and open-air locations like Bar 25 or the Club der Visionäre (CDV) became internationally prominent and were used as sceneries for numerous movies and novels. Together with a number of additional clubs (Tresor, 103 club), open-air bars (YAAM, Kiki Blofeld, etc.), labels (e.g. Groove), and record stores (e.g. Hardwaxx) these locations achieved a legendary character even on an international scale, attracting a new form of party tourism, which the literature has named "*easyjetset*" (Rapp, 2009).[3] In sum, the area represents one of the most important music clusters within a city that is – as it is popularly associated in newspapers and magazines – often perceived as *the* city of contemporary music (van Heur, 2009).[4]

In 2002, the Universal concern moved its European headquarter from Hamburg into the area, exactly vìs-a-vìs the Watergate club, separated only by the river Spree. Two years later MTV Europe followed, choosing a riverbank location just a few hundred meters away. Both enterprises received public support running into the millions, but the culturally fertile environment represented an asset in terms of location attractiveness. These enterprises should have served

[1] A number of recent empirical studies suggest that, in practice, displacement of low-income households only occurs to a relatively limited degree (Freeman and Braconi, 2002; Vigdor, 2002). Newman and Wyly (2006), however, argue that such figures usually underestimate displacement due to neglecting households that left cities, doubled up with other households, became homeless or entered the shelter system.

[2] See e.g. Lees (2000) for a survey.

[3] Note that during the study period the airport Berlin-Schönefeld was one of the major hubs of the low-cost carrier "easyJet".

[4] Van Heur provides a comparative analysis of contemporary music nodes in London and Berlin.

as anchor users within an area which was chosen to become one of Berlin's largest investment projects with a particular focus on communication and media industries, the so-called Mediaspree. The project, which was promoted in a public–private partnership by private investors, authorities and the local chamber of commerce, had been under discussion since the mid-1990s, but became concrete after 2002 when the Senate Department adopted the land-use plan in order to facilitate the renovation of old warehouses and the development of empty properties to be used for offices, lofts, hotels and a new 17,000-seat multifunctional event arena completed in 2008. The designated area covers about 180 ha along approximately 3.7 km of both riversides. The prestigious project had been intended to bring positive economic impulses to the area and to contribute to the revitalization of the surrounding south-eastern downtown areas, which were found to be in need of external stimuli. It has been embedded into areas that belong to two major German urban redevelopment programs Stadtumbau Ost and Stadtumbau West, which have been running since 2002 on the Friedrichshain side and 2005 on the Kreuzberg riverside. These programs involve considerable investment into public infrastructure as well as monetary incentives for private investment in housing stock renewal. Generally, the plans share many similarities with international urban regeneration policies such as those stated in the UK Urban Task Force *Towards an urban renaissance* or the U.S. Department of Housing and Urban Development *The state of the cities* reports (DETR, 1999; HUD, 1999), putting issues like the provision of attractive public spaces, sophisticated architecture and urban design as well as the integration of arts and media on the planning agenda. Fig. 1 shows the designated Mediaspree area and the adjacent redevelopment areas as against the background of the local music geography.

The stated objective of authorities to induce economic stimuli and increase the attractiveness of the location, accompanied by the creation of luxury housing, retailing, office space and mainstream entertainment, raised fears and anger among the resident population around the Mediaspree area. The typical quarrel flared up, running along the lines that one man's gentrification is another man's displacement (Newman and Wyly, 2006). According to the stage-model, such conflicts typically occur at the third stage, when the first-stage pioneers run out of affordable living space due to the immigration of higher income gentrifiers to whom the area becomes increasingly attractive after the initial neighborhood change (Friedrichs, 2000). Along these lines, the key arguments of the opponents of the Mediaspree project focused on a threat to the urban equilibrium due to an increase in the cost of living space and subsequent displacement pressures. It was also argued that

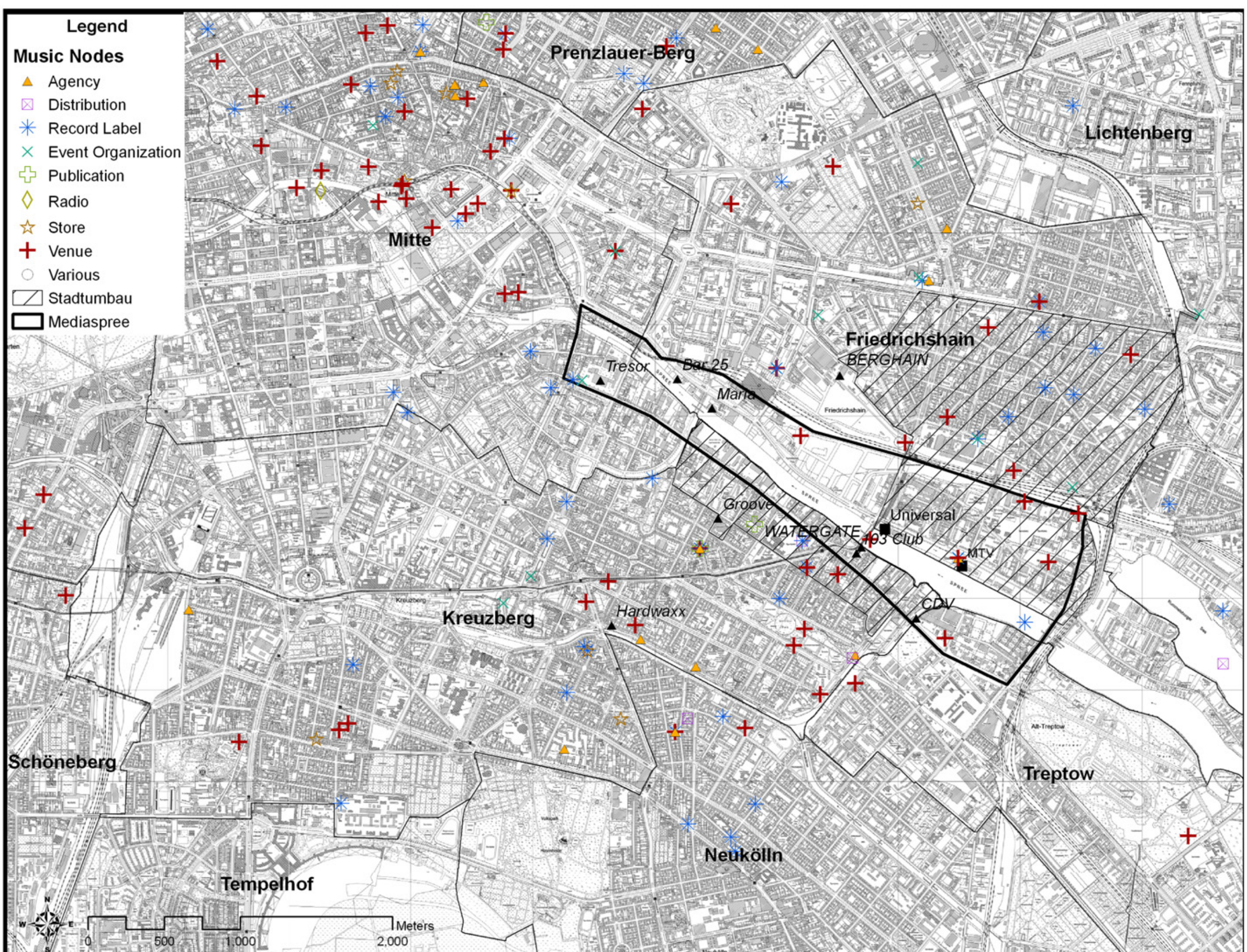

Fig. 1. Neighborhood situation. Notes: The figure has been created on the basis of the urban and environment information system (Senatsverwaltung für Stadtentwicklung Berlin, 2006). Locations of music nodes have been geocoded based on van Heur (2008).

the urban renewal would leave no room for the unique cultural diversity within the area (Lee and Hebel, 2007). To comprehend these fears, the background of the extremely low rate of owner occupancy within the study area has to be understood. In 2006, 139,200 from a total of 145,530 apartments were rented out, which is equal to more than 95% (Investitionsbank Berlin, 2007). This is a very high figure, even compared to most of the other districts in Berlin. Any increase on the demand-side capitalizing into rent-levels therefore increases the revenues of landlords and the cost of living for almost all residents in the area.

Note that the Mediaspree area itself has been very sparsely developed and the project could therefore hardly displace population within that area. This is of course not the case for the adjacent revitalization areas. Moreover, changes within the development as well as the redevelopment areas were expected to affect the neighborhood beyond the fairly administrative boundaries. The Mediaspree project, therefore, served as a catalyst for perceived threats of neighborhood change and gentrification among the residents, which were certainly partially, but not entirely, attributable to the project itself. Driven by the activist groups "*AG Spreeufer*" (Spree riverside) and "*AG Spreepiratinnen*" (Spree pirates) a citizens' initiative was formed called "*Mediaspree Versenken*" (Sink the Mediaspree), which advocated for a reduction of building density and height and the preservation of public space along the riverbanks to be used as a subsidiary for recreational and cultural activities. Within a period of five months with numerous demonstrations and protest events, the initiators had collected enough signatures to enforce a public referendum against the Mediaspree plans. The referendum, which had no binding character, was held on July 13, 2008, under the label "*Spreeufeer für alle*!" (Spree riverside for all!). Residents were asked whether they would support a ban for buildings exceeding 22 m heights (the so-called "Traufhöhe") and the creation of a public riparian strip of at least 50 m width. The referendum won the approval of as much as 87% of the voters, with a turnout that was significantly above the necessary threshold (19.1% vs. 15%). A counterproposal from the district authority was in no position to win a majority. Briefly summarized, the public protests and the clear majority vote reflect that a large number of residents were in strong opposition to the project. The most important arguments were about the destabilization of the neighborhood (threat of displacement and loss of cultural diversity) while criticism of the economic viability of the projects, e.g. the superfluity of an additional multifunctional sports arena in Berlin, was existent, but not the focus of the resistance.

3. Empirical analyses

3.1. Precinct level voting analysis

Besides a huge body of theoretical literature, the complex phenomenon of gentrification has also been approached empirically from various disciplinary perspectives (e.g. Brueckner and Rosenthal, 2005; Freeman and Braconi, 2002; Knotts and Haspel, 2006; McKinnish et al., 2008; Vigdor, 2002, among many others). Quantitative techniques are frequently employed to explain the objective measures of displacement in terms of characteristics of the renters. Qualitative analysis based on interviews instead may reveal which of the residents still living in the neighborhood face a perceived threat of displacement and neighborhood change. Newman and Wyly (2006) provide an interesting comparison of results based on both techniques. Our empirical strategy in some sense represents a hybrid of both approaches. The empirical analysis of the precinct level voting outcome of the Mediaspree referendum, given that it was regarded as a major driving force of gentrification, facilitates inference on population groups that particularly opposed the ongoing and expected neighborhood change. The clear limitation of this approach is the impossibility to account for individual characteristics in the same detail as in qualitative interviews. The straightforward advantage, however, is the possibility to evaluate the opinions of ten-thousands (or more) of residents, which – under reasonable constraints – can hardly be achieved in field studies. Our approach may therefore be regarded as complementary to the established qualitative techniques.

From the 34,326 valid votes in the referendum, 27,667 votes from the ballot boxes can be utilized in the empirical analyses. The remaining absentee votes, unfortunately, cannot be considered due to missing geo-references. Data are available at the precinct level from the district authority District Authority Friedrichshain-Kreuzberg (2008). At city level, there are 1201 voting precincts, 87 of which are within our study area comprising the city district Friedrichshain–Kreuzberg. We use a GIS framework to merge the voting outcome with 2008 data on socio-demographic characteristics available at the levels of 15,937 statistical blocks (population, age groups, proportion of male and non-German population), 191 zip codes (purchasing power) and 2424 small voting precincts (outcome of the 2006 state elections). These data were obtained from the Statistical Office in Berlin, with the exception of data on purchasing power, which was originally derived from a prognosis of the consumer research society (Gesellschaft für Konsumforschung (GfK)). Standard area interpolation techniques (Arntz and Wilke, 2007; Goodchild and Lam, 1980) are used to aggregate all data to the level of the 87 precincts within our study area.

It is important to note that due to the non-binding nature of the referendum, there were asymmetric incentives to engage in the public vote for opponents and proponents in the referendum. Those who decided to engage in the referendum are likely to be a sample selected a group of the underlying population. As is shown in Fig. A1 in the appendix, the rejection rate varied as few as between 84 and 87% along all buffer distances to the treatment area. Fig. A1 also shows that the spatial variance in opposition is almost entirely driven by turnout, which implies that participation was severely biased towards opponents of the project. An *ecological inference* on the individual behavior of representative residents based on the precinct level voting outcome is, hence, likely to lead to an *ecological fallacy* and is not appropriate in this case.[5]

We therefore depart from the common strategy in the applied public choice literature, which from a comparison of yes- to no-votes infers to the behavior of a representative (median) voter. Instead, we use a proportion of yes-votes from the number of eligible voters as our primary dependent variable. Rather than inferring to individual effects, we interpret this variable as a proxy of neighborhood unrest within a given precinct. We assume that larger opposition against the project within a precinct strictly increases the proportion of residents supporting the referendum and, ergo, the number of yes-votes relative to the number of eligible voters. Not surprisingly, unpublished robustness checks using turnout as a dependent variable yield almost exactly the same results throughout all stages of the empirical analysis. The spatial distribution of our primary indicator variable is depicted in Fig. 2.

The picture clearly indicates a high degree of variation in the mobilization of Mediaspree opponents, ranging from as few as 2% to more than 40% of the elegible voters. It is also evident that almost all of the precincts with a mobilization rate of more than 15% lie within a 1 km buffer area surounding the Mediaspree area, indicating a relatively localized opposition to the project. This is a fairly intuitive finding, given that the perceived threats associated with the project should be largest in the immediate proximity. Both the expected impact on affordability of living space as well as the disamenity effect

[5] Instead of individual data, the preferences of the inhabitants can also be examined on the basis of aggregated or grouped population statistical data, e.g. at the constituency level. An extensive discussion of the underlying assumptions of ecological inference can be found in Shively (1969), King (1997), or King et al. (2004).

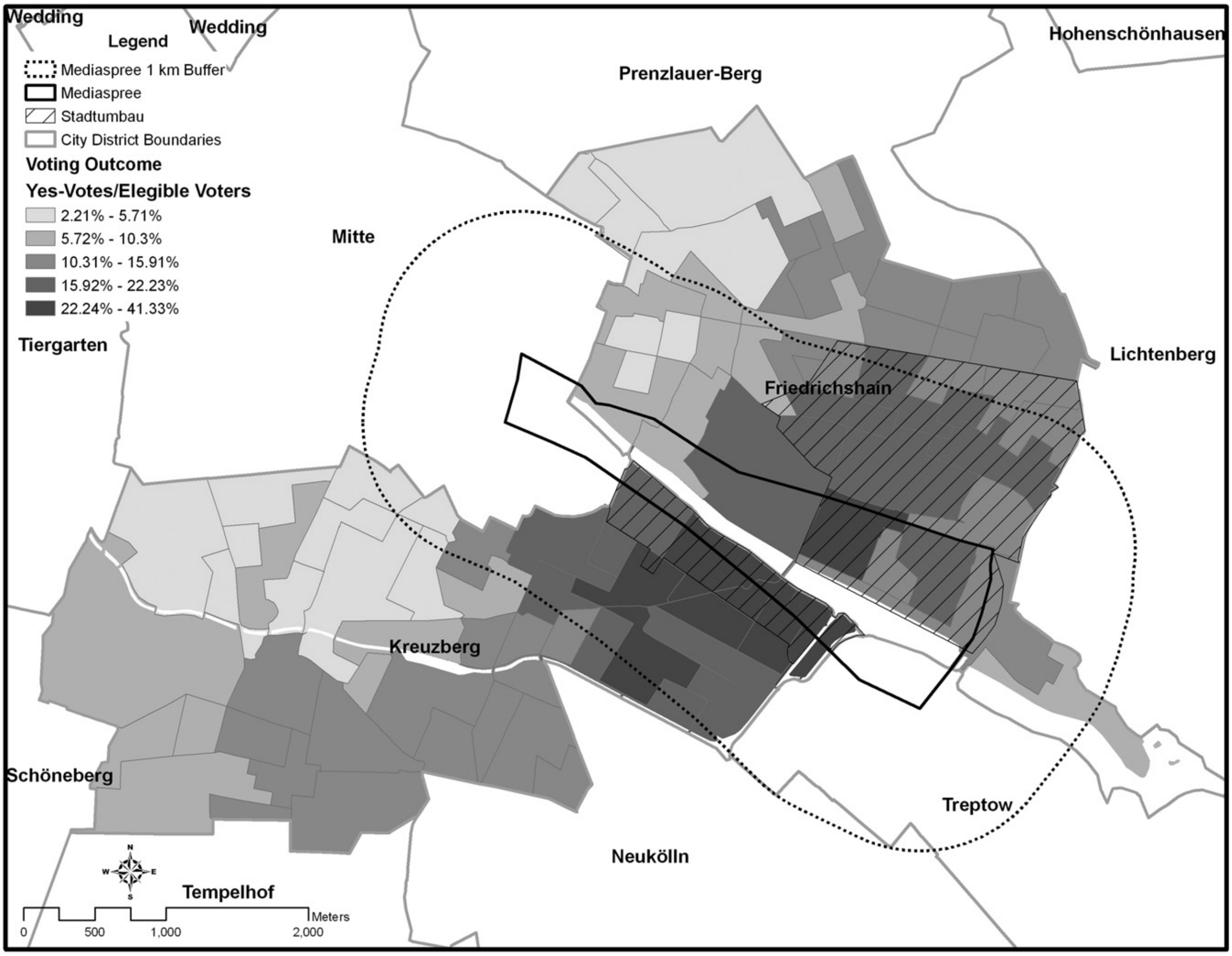

Fig. 2. Voting pattern. Notes: This figure has been created on the basis of the urban and environment information system (Senatsverwaltung für Stadtentwicklung Berlin, 2006). Classes are defined according to the Jenks (1977) algorithm.

of the neighborhood change along the riversides are likely to diminish with distance.

We employ a strategy that is well-established in the applied public choice literature (e.g. Brunner et al., 2001; Coates and Humphreys, 2006; Dehring et al., 2008; Rushton, 2005) in order to reveal the precinct characteristics that impact significantly on the voting outcome and to observe whether there is any proximity effect conditional on these characteristics. The following specification is used:

$$PctYES_i = \alpha + \sum\nolimits_k \beta_k X_{ik} + \gamma_1 west_i + \gamma_2 revWest_i + \gamma_3 revEast_i + \gamma_4 distMS_i + \varepsilon_i \quad (1)$$

where $PctYES_i$ is the percentage of yes-votes of eligible voters in precinct i, ε is the error term and the other Greek letters are coefficients to be estimated. The socio-demographic characteristics of precincts are represented by a set of variables X_k. An anticipated increase in the cost of living space should particularly affect those residents that face relatively tight budget constraints. From this, a potentially stronger opposition against the Mediaspree project might arise in precincts with a relatively low-income level, which we approximate using *purchasing power* per capita. Since the cultural establishments discussed above mainly address a relatively young

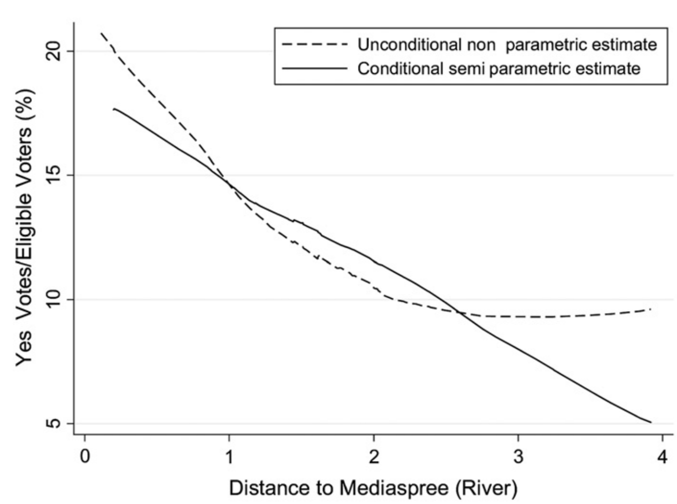

Fig. 3. Non-parametric distance effects. Notes: Gradients are estimated by use of locally weighted regressions. Conditional estimates are obtained employing the Lokshin (2006) technique.

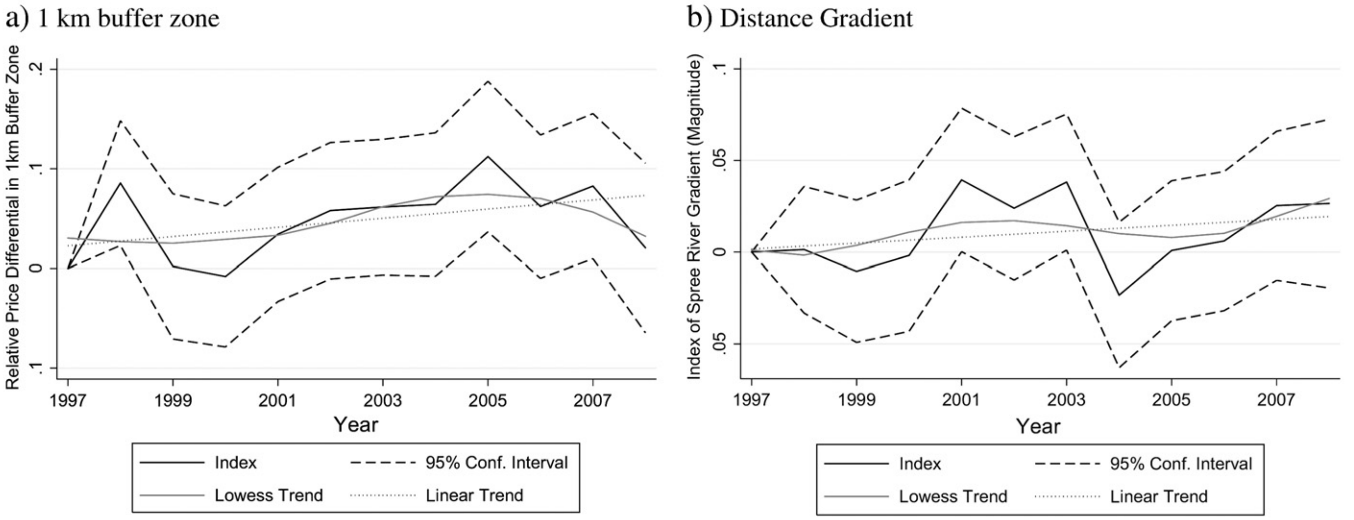

Fig. 4. Indices of relative apartment prices. Notes: Figures generated on the basis equation (2). Estimation results for control variables are pre-sented in Table A1, (1) and (2) in the appendix. The Lowess trend is a non-linear fit using lo-cally weighted regressions.

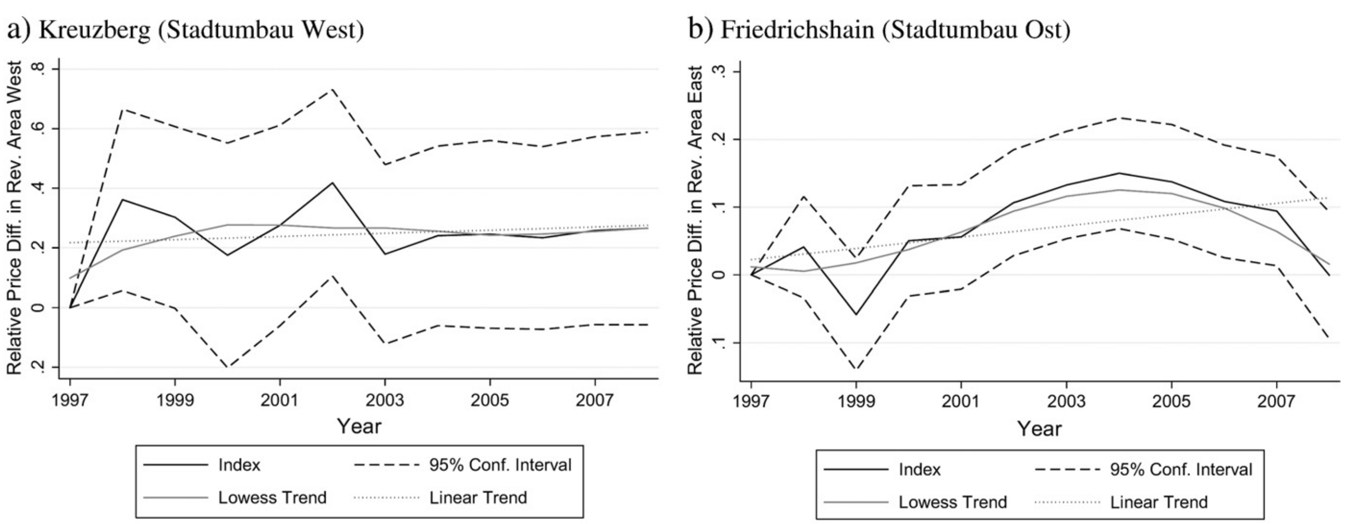

Fig. 5. Indices of relative apartment prices. Notes: Figures generated on the basis equation (2.) Estimation results for control variables are pre-sented in Table A1, (3). The Lowess trend is a non-linear fit using locally weighted regres-sions.

adult audience, we add the proportion of 18–45 year-olds, who potentially expect a larger (dis-)utility from a loss of these amenities compared to other population groups. Since the (expected) utility may also vary across other population groups, we add the proportion of *male* and *non-German* population. The latter variable may also pick up the effect of social and language barriers, which might lead to lower mobilization in precincts with a high proportion of foreigners.

As an additional set of controls, we use political party affiliation in the 2006 state election, mainly for two reasons: First, the project was supported by the state government, which could impact on the attitudes of supporters of governing and opposing parties. Second, political party affiliation serves as a rough proxy for lifestyle-group belonging, which is not entirely captured by standard socio-demographic attributes. As noted in the death-of-class-debate a one-dimensional view on society along a linear income-ray falls short in accounting for the full diversity of personal tastes, attitudes and values, as well as political orientation, and consumption preferences (Veal, 1993). Lifestyle particularities may well determine the subjective benefits from the local cultural amenities. Last, we use the precinct level *turnout* in the 2006 federal election as a proxy for the general political interest, which may affect the likelihood of residents engaging in the subject referendum.[6]

The remaining variables in specification (1) are of a geographic nature, denoting whether a precinct lies on the Kreuzberg side of the river (*west*), within the area of the revitalization program Stadtumbau West (*revitalization west*/*revWest*) or Stadtumbau Ost (*revitalization east*/*revEast*) and the distance from the precincts' centroids to the

[6] The 2006 turnout within the study area was in line with the rest of Berlin (59.5% compared to 60.5%) and that we cannot reject that the mean turnout in the referendum did not vary between the 1 km buffer zone and the rest of our study area based on a *t*-test.

Fig. 6. Precinct level appreciation rates. Notes: Figure has been created on the basis of the urban and environment information system (Senatsverwaltung für Stadtentwicklung Berlin, 2006). Classes are defined according to the Jenks (1977) algorithm.

respective riverbank alongside the Mediaspreeareal (*distance to river/distMS*). These variables are introduced into the specification in order to test the hypothesis of no spatially uneven resistance, conditional on precinct characteristics.

Estimation results for specification (1) are presented in Table 1. In column (1) the hypothesis is tested that support for the referendum is spatially homogenous, unconditional on precinct characteristics. This hypothesis has to be rejected as suggested by Fig. 2. Support is generally larger within Kreuzberg than Friedrichshain and within both revitalization areas compared to the rest of the district. Furthermore, support significantly diminishes with distance to the riverbanks. In column (2) the precinct characteristics discussed above are introduced into the model in order to test the same hypothesis, conditional on socio-economic attributes. After stepwise deletion of insignificant variables the final specification (3) is obtained. Socio-demographic characteristics explain much of the differences between the average support in Friedrichshain–Kreuzberg as well as between the revitalization areas and the rest of the study area, but not the significant impact of proximity to the riverbanks. Accordingly, the proportion of yes-votes of eligible voters decreases by about 3.5 percentage points for any 1 km increase in distance to the riverbank.

From the socio-demographic characteristics, a number of variables are found to have robust and significant effects. An increase in the proportion of relatively young adults by 1 percentage point yields a 0.31 percentage point increase in the rate of approval. A similar increase in the proportion of supporters of the left-hand side governing parties (*Gov* is formed by *SPD* and *Die Linke*) as well as of the major conservative opposition party (*CDU*) in the 2006 state elections induces a decrease in the rate of approval by about 0.25 and 0.57 percentage points.[7] This effect could be either attributable to voters feeling committed to the plans of the parties to which they are affiliated (all of the three parties supported the project) or to lifestyle-specific attitudes and the tastes of supporters of these mainstream parties, which may have a relatively lower demand for alternative cultural goods. The turnout in the federal elections, in contrast, impacts positively on the support of the referendum. Precincts with a high proportion of politically active residents also exhibit a larger rate of mobilization and approval in the subject referendum. The increase in the coefficient of determination of about 0.5 in columns (2) and (3) compared to (1) indicates that these socio-demographic precinct characteristics account for a relatively large proportion of the variation in the voting outcome. From the socio-economic variables that do not exhibit a significant impact on spatial voting pattern *purchasing power* is the most notable as a significantly positive impact

[7] Individually, the governing parties SPD (−0.24) and Die Linke (−0.26) exhibit almost exactly the same coefficient values.

Table 1
Determinants of yes-votes/eligible voters.

	(1) OLS	(2) OLS	(3) OLS	(4) SAR
West (dummy)	4.924**	5.077	1.869	1.823
	(1.861)	(4.347)	(1.883)	(1.845)
Revitalization west (dummy)	7.196*	2.049	2.531	2.598
	(3.153)	(2.335)	(1.980)	(2.021)
Revitalization Ost (dummy)	6.575**	0.144	0.124	0.137
	(1.307)	(1.209)	(1.124)	(1.109)
Distance to river (km)	−3.052**	−3.719**	−3.498**	−3.371**
	(1.074)	(0.928)	(0.694)	(0.923)
Purchasing power (€/capita)		0.0003		
		(0.001)		
18–45 year-olds (%)		0.176*	0.131*	0.128*
		(0.075)	(0.051)	(0.052)
Non-German (%)		−0.095		
		(0.095)		
Male (%)		−0.075		
		(0.180)		
CDU (%)		−0.611**	−0.569**	−0.554**
		(0.085)	(0.071)	(0.086)
Government parties (%)		−0.205*	−0.245**	−0.238**
		(0.079)	(0.073)	(0.068)
Turnout (%)		0.119	0.205*	0.210**
		(0.108)	(0.080)	(0.075)
Constant	13.585**	27.541*	18.106*	17.013*
	(1.727)	(16.431)	(9.124)	(8.620)
Rho				0.005
				(0.014)
Observations	87	87	87	87
(Pseudo) R-squared	0.38	0.85	0.85	0.85
Mean VIF	1.39	6.52	4.27	4.27
AIC	533.83	442.61	438.95	442.69

Notes: Endogenous variable is percentage of yes-votes at eligible voters in all models. Robust standard errors in parentheses.
** $p<0.05$.
* $p<0.01$.

would have been expected if affordability issues were the major concern of opponents.

A critical question regarding the specification of model (3) is whether the distance to the riverbanks should be assumed to impact linearly. We therefore estimate the unknown non-linear relationship between the percentage of yes-votes of eligible voters and the distance to the river $f(distMS)$, both unconditionally as well as conditionally on controls used in Table 1, column (3).

$$PctYES_i = \alpha + \sum_k \beta_k X_{ik} + \gamma_1 west_i + \gamma_2 revWest_i + \gamma_3 revEast_i + f(distMS_i) + \varepsilon_i \quad (2)$$

Following a tradition established in important contributions such as Cleveland and Devlin (1988) and McMillen (1996), the relationship is estimated by means of locally weighted regressions. Non-linear estimates of the impact of distance on the voting outcome are visualized in Fig. 3 for a semi-parametric as well as a reduced non-parametric version of Eq. (2). While unconditional estimates indicate a marginal impact that diminishes with distance, the relationship conditional on Table 1, (3) control variables exhibits a fairly linear shape, suggesting that the specification employed in Table 1 is appropriate.

Another important issue regarding the appropriate spatial specification of model (3) is the presence of spatial dependency, which can cause inefficient or biased OLS estimates. Based on a row-standardized contiguity weights matrix, however, LM-tests do not reject the hypothesis of no spatial autocorrelation.[8] If a spatial lag model is employed (Eq. (4)) in order to account for a potential dependency of the precinct outcome on neighboring outcomes, which might result from cross-border interactions between voters, coefficient estimates remain almost unchanged while the spatial lag coefficient (Rho) remains statistically insignificant.[9] Spatial dependency, hence, does not seem to give much cause for concern.

It is noteworthy that the official plans ensure public access to the riversides, which will even be improved considerably on the Kreuzberg side of the river. It is therefore unlikely that we are observing an (expected) disutility effect related to the accessibility of the river Spree as a natural amenity. There are, however, at least two competing hypotheses that can be developed on the basis of the arguments of the citizens' initiative and the results presented above: First, the perceived disutility of losing the cultural amenities along the riverbanks should diminish with distance. The net-utility of those amenities will generally decrease with transport costs, leading to lower incentives to engage for their preservation at larger distances. Closely related, we may observe the effect of a Tiebout-like (1956) sorting process with respect to the local cultural amenities and residents' preferences and tastes. Unobserved residents' characteristics may well account for the otherwise unexplainable localized support in the referendum. Coates and Humphreys (2006) develop a similar argument as an explanation for the local support for professional sports facilities. Waldfogel (2008) provides empirical evidence for a local matching of local private goods and residents' taste. Second, the concerns of raising the cost of living space due to the project should be larger at close distances, given that the spillovers are likely to be localized. We will turn our attention to this point in the next sub-section.

3.2. Local apartment price appreciation

Ambitious planning authorities aiming to revitalize downtown areas have – at the very least – an ambiguous if not different perspective on gentrification and area valuation compared to local renters. Besides a general concern about the social cost for the displaced, the gentrification process might be regarded positively or negatively for the residents avoiding displacement. On the one hand, residents may benefit from neighborhood improvements while, on the other, they suffer from the displacement of culture and community networks (Atkinson, 2000; Marcuse, 1986). Anyway, authorities that aim at improving neighborhood quality will have to acknowledge that an increase in the attractiveness of places leads to a rise in the willingness to pay for living space, if not by the old inhabitants then by those who decide to move into the neighborhood. A demand-side driven increase in the price of living space may, in this light, even serve as a benchmark for the success of revitalization attempts.

The assessment of the Mediaspree development project and the two Stadtumbau (Ost and West) redevelopment projects in these terms, however, is complicated by the fact that the urban intervention did not come on one identifiable date. Standard difference-in-difference approaches or regression discontinuity designs compare appreciation within a treatment relative to a control area before and after an a priori known intervention. As noted above, most of the plans for the Mediaspree date back to the 1990s, but materialized only gradually due to the bad economic climate. While the development gained some pace after 2002, when the Senate Department adopted the land-use plan, effects may have well been anticipated by real estate markets before (McMillen and McDonald, 2004). We therefore employ a flexible specification that allows treatment effects to vary over a sufficiently long period. If the project had the potential to emanate positive externalities on the attractiveness of the area, its gradual evolution together with the decreasing uncertainty about the outcome would be mirrored in rising (relative) real estate prices.

[8] The LM-test scores are: LM_{error}: 0.244, LM_{lag}: 0.240.

[9] An alternative form of spatial dependency would result from spatial measurement errors or omitted variables that are correlated across space. This form of spatial dependency can be dealt with a spatial error correction model. Methodological background to spatial lag and spatial error models are covered by Anselin (1988), among others.

As noted above, Berlin in general and the study area in particular are characterized by an extremely low rate of owner occupancy. The effective cost of living space is therefore determined by the local rent level. Housing prices, however, are just a reflection of expected (discounted) revenues of investors, which depend on realizable rents. While rents due to rigidities and legal constraints adjust gradually, shocks to changes in demand should capitalize immediately into house prices. We will investigate the evolution of house prices within the study area based on all apartment transactions that occurred from January 1997 to June 2008, the month before the referendum was hold. Our data set, obtained from the local Committee of Valuation Experts in Berlin (2008), contains many of the usual features (e.g. age, size, number of rooms, and balcony) and, in addition, some additional contract details (e.g. tax privileges, rent guarantees). After losing a handful of observations due to missing values we obtain a final sample of 9980 transactions. Data are merged with the precinct level framework within a GIS environment that also facilitates the calculation of environmental variables.

We adopt a hedonic approach using the well-established log-linear specification in order to correct for apartment as well as location characteristics. This approach is in line with a large body of literature that shares the idea of treating real estate commodities as bundles of attributes whose implicit prices can be estimated using multivariate regression (Rosen, 1974). Using log of prices per square meter (*Psqm*) as the dependent variable, our regression specification basically takes the following form:

$$\log\left(Psqm_{jt}\right) = \sum_m \vartheta_m Y_{jm} + \delta Z_j + \sum_{1998}^{2008}\left(\delta_u Z_j \times \phi_t\right) + \phi_t + \varphi_i + \omega_j \tag{3}$$

where Y_m are the structural and locational control variables listed in Table A1 in the appendix and ϑ_m the respective estimated marginal price effects. Parameters Φ_t and φ_j represent full sets of time and precinct effects and control for unobserved location characteristics and macro-economic shocks that affect the entire sturdy area, while ω_j is an error term. Z describes the location of a property transaction with respect to the (re)development areas of interest. In the first step Z is a dummy variable denoting transactions that occurred within the 1 km buffer zone depicted in Fig. 2. Coefficients δ_u form an index of relative prices within that area relative to the initial year 1997. In some sense they provide difference-in-difference estimates as they differentiate (conditional) mean prices over space and time. Column (1) in Table A1 in the appendix shows estimates using an established set of structural attributes supplemented by location controls such as the distance to the central business district (*CBD*), the nearest *school*, *park*, *water* body and metro *rail station*.[10] These variables account for transport costs to these features that are traded against the price of living space. Recent research indicates that the historic quality of a neighborhood's building stock may also represent a valuable location amenity (e.g. Ahlfeldt and Maennig, 2010; Coulson and Lahr, 2005), which we address by the distance to the nearest designated *landmark*. With only a few exceptions, the coefficient estimates are in line with conventional expectations and recent evidence for the Berlin housing market (Ahlfeldt, in press). Notable is the positive coefficient on *distance to CBD*, probably revealing that the amenity effect of peripheral recreational amenities dominates the centripetal forces, e.g. labor market accessibility.

Estimation results for δ_u are presented in Fig. 4a, together with the respective 95% confidence interval and linear and non-linear trend lines. Note that in Figs. 4 and 5 the non-linear trend lines are generated as a smoothed function of the estimated time-varying treatment parameters. McMillen and Thorsnes (2000;2003) show how, in an alternative approach, a corresponding non-linear trend line may be estimated using semi-parametric regression techniques.

If any, Fig. 4a shows only a weakly positive long-term trend in prices within the 1 km buffer zone relative to the rest of the district. While there is a steady increase up to 2005, a pronounced downward adjustment occurs afterwards. This depreciating effect might be related to the construction of the o2 arena, a 17,000 seat multifunctional sports area, as professional sports facilities may induce perceived (expected) proximity cost related to noise and congestion (Ahlfeldt et al., 2010). The year 2005 is the only year where the relative price differential can be rejected to be zero on the basis of the 95% confidence interval. Fig. 4b provides analogous estimates with Z representing the distance to the Mediaspree riverbanks as used in Table 1. Estimates for control variables are provided in Table A1, (2). Note that Fig. 4b shows the magnitude of the coefficient so that positive values imply an increase in the marginal price effect of 1 km distance. Again, there is only, if any, a weakly positive evolution of land prices with respect to proximity to the treatment area. A positive trend up to 2003 is offset by pronounced adjustment afterwards, followed again by a positive development after 2004. Moreover, the gradient cannot be statistically rejected to take the same value as in the initial period in all years.

As noted, it is difficult to identify a clear intervention date for the Mediaspree development project itself. For the adjacent urban redevelopment areas, however, such intervention dates can easily be identified. If the redevelopment programs had led to an increase in attractiveness of the neighborhood and a corresponding increase in demand for living space, we would expect a positive reaction in relative prices after 2002 in the case of the eastern Friedrichshain riverside and a respective adjustment within the western Kreuzberg side after 2005. Again, we employ an Eq. (3) type specification where Z now represents a vector of two dummy variables which each denote one of the two redevelopment areas. Results for control variables are presented in column (3) of Table A1 while indices of relative land prices are shown in Fig. 5. Fig. 5a do not point to any significant relative price trend within the western redevelopment area at all. Either investments were not large enough to trigger significant price effects or the effects are not visible, yet. In contrast, Fig. 5b reveals significantly positive price differentials within the eastern redevelopment area between 2002 and 2007, with the first significant year being the implementation year of the redevelopment program. The positive trend, however, seems to start before 2002 and does not hold longer than 2005. In the subsequent period prior to the referendum relative prices declined until the previous increase is almost offset.

In our last approach to the evaluation of apartment prices within the study areas we turn our attention to individual trends at the precinct level on the basis of the following specification:

$$\log\left(Psqm_{jt}\right) = \sum_m \vartheta_m Y_{jm} + \sum_i \lambda_i TREND_t \times \varphi_j + \phi_t + \varphi_j + \upsilon_j \tag{4}$$

where φ_i again is a set of precinct level dummies. These are also interacted with a quarterly time $TREND_t$, which is rescaled in a way that λ_i coefficients give the average yearly appreciation in apartment prices at precinct *i*. As described in more detail in the next section, the idea is to investigate whether precinct level appreciation significantly explains the voting pattern in the referendum. Since a priori the period over which residents perceive appreciation in their neighborhood is not clear, we generate a set of trend estimates, each one starting at a different year from 1997 to 2007 and opt for a starting date in 2000 based on the Akaike information criterion obtained in Eq. (4). Baseline estimation results corresponding to Eq. (4) are presented in column (4) of Table A1. The

[10] The CBD is defined as the crossroads between the boulevards Friedrichstrasse and Leipziger Strasse.

estimated trend effects, which are almost all statistically significant at conventional levels, are visualized in Fig. 6. From the picture, however, no comprehensive spatial pattern is immediately apparent. Note that we lose 9 precincts due to insufficient transactions to establish an individual precinct trend estimate.

From the results presented in this section we cannot affirm that property owners acknowledge or expect any significant impact on the location desirability from the Mediaspree project. While this may be a disappointing result for authorities fostering an increase in neighborhood quality, it is encouraging for the resident population, given that at least the plans don't seem to amplify displacement pressures.

3.3. Voting pattern, appreciation and amenities

As noted above, activists engaging in the resistance against the Mediaspree project were worried about an increase in costs for living space in the area on the one hand, and about the loss of vivid public and cultural spaces along the Spree riverbank as well a change in neighborhood character on the other. These concerns stand exemplarily for residents in gentrifying neighborhoods fearing for their own displacement or the displacement of culture and community networks (Atkinson, 2000; Marcuse, 1986). As shown in the section above, evidence that the major urban development and redevelopment projects in the area have had a significant impact on apartment prices is hardly compelling. Nevertheless, there is considerable heterogeneity in appreciation rates across precincts as indicated by Fig. 6. These are a good predictor for the relative evolution of rent-levels since buyers anticipate future revenues in their bids. If residents in precincts with relatively higher appreciation rates perceive higher displacement pressures, we would expect them to be more forceful in opposing a project that has been argued to increase demand for living space in the neighborhood by both proponents (authorities) and opponents (citizens' initiatives).

The economic rationale is straightforward. In principle, we may assume that the established residents had chosen an optimum neighborhood to live in, trading the neighborhood quality against the cost for living space, conditional on their budget constraints. In this situation an increase in neighborhood quality (due to an exogenous urban development project) will attract gentrifying households who are prepared to pay higher marginal prices for neighborhood quality, raising the local rent level. The old inhabitants are thereby driven out of their consumption optima. The positive utility effect of an increase in neighborhood quality is overcompensated by reduced non-housing consumption. Residents who decide to stay and to engage with increasing rents are forced to consume too much neighborhood quality and too few non-housing goods compared to the optimum allocation. The resulting disutility comes in addition to the disutility from an adjustment of local services and cultural amenities in favor of the gentrifiers. Given that there is a minimum consumption of non-housing goods, displacement pressures – at least theoretically – can eventually become high enough to leave residents with no choice but to exit their neighborhood. This scenario corresponds to what Marcuse (1986) calls exclusionary displacement. It is worth noting that this rationale is in contrast to the implications of the homevoter hypothesis, which is supported by empirical research for the U.S. (Brunner and Sonstelie, 2003; Brunner et al., 2001; Dehring et al., 2008; Hilber and Mayer, 2009). Accordingly, homeowners will vote in favor of public goods or any kind of initiatives that they expect will raise the value of their immobile assets — their real estate properties (Fischel, 2001).

If we assume residents to perceive the experienced local relative appreciation as a (noisy) signal for the impact of the ongoing Mediaspree and Stadtumbau Ost/West (re)development projects on local demand for living space in the broad area of Friedrichshain–Kreuzberg, inference on the validity of the homevoter hypothesis within rental environments is possible on the basis of the conditional relationship between support for the subject referendum and local appreciation rates. Therefore, we introduce the estimated local appreciation rates $\hat{\lambda}_i$ from Eq. (4) into specification (1).

$$PctYES_i = \alpha + \sum_k \beta_k X_{ik} + \gamma_1 west_i + \gamma_2 revWest_i + \gamma_3 revEast_i + \gamma_4 distMS_i + \eta\hat{\lambda}_i + \tau DuM_i + \varepsilon_i \quad (5)$$

Since the introduced variable is a generated regressor, similarly to Dehring et al. (2008) we bootstrap standard errors in 500 replications in order to avoid potential bias in OLS standard errors (Murphy and Topel, 2002). As noted above, appreciation rates could not be estimated for a handful of precincts. In these cases, appreciation rates are set to zero and the respective observations denoted by a dummy (*DuM*).

Results corresponding to this specification are presented in Table 2, (1). They show that, as expected, the homevoter hypothesis in its standard form does not apply to the subject market with marginal owner occupancy. Besides the clear opposition against the potentially price appreciating project at district level – at least by those who engaged in the referendum – we find a significantly positive relationship between *appreciation* and opposition across precincts, although it is relatively moderate. Accordingly, an increase in the average appreciation during 2000–2008 by 1 percentage point leads to a 0.12 percentage point increase in the proportion of voters supporting the referendum, corresponding to about a 1% increase relative to the rate of approval at the district level (13.1%). Potentially, voters in the proximity of the Mediaspree riverbanks associate more strongly a perceived appreciation within their neighborhood with the project. Since we then would expect a relatively larger marginal impact on the voting outcome, we allow the marginal impact to vary with distance to the Mediaspree by the introduction of an interactive term *DistMS*×*Appreciation* in column (2). While the coefficient shows the expected sign, standard errors by far exceed a threshold that would be required to satisfy conventional significance criteria.

One explanation of the relatively modest effect of the appreciation variable could be that, despite the included neighborhood controls, there are unobserved preferences, e.g. a more conservative orientation, which are correlated with appreciation rates and downward bias the respective estimates. Such a correlation could likely emerge from an inflow of new residents who have fewer reservations against the ongoing neighborhood change. We do not control for migration in the precinct level regressions as data are only available for the next higher level of aggregation, which are traffic cells of which only 16 lie within our study area. If we aggregate the estimated appreciation rates to traffic cells to establish the correlation with the mean ratio of total migration at total population in 2002–2006, we find a coefficient that is not statistically distinguishable from zero (−0.037), which somewhat alleviates the related concerns.[11]

Note that spatial dependency, again, does not appear to represent a major concern. Based on a row-standardized contiguous weights matrix the LM-test cannot reject the hypothesis of no spatial dependency. The application of a spatial lag model (Eq. (3)) leaves the estimates almost unchanged while the lag coefficient itself is not statistically significant.[12]

[11] We note that due to the very low home ownership ratio with very limited variation throughout the study area it is not feasible to assess the impact of appreciation on the voting behavior of homeowners and renters separately.

[12] The LM-test scores are: LM_{error}: 0.027, LM_{lag}: 0.152.

Table 2
Impact of apartment appreciation on yes-votes/eligible voters.

	(1) OLS	(2) OLS	(3) SAR
West (dummy)	1.442	1.288	1.448
	(1.653)	(1.736)	(1.590)
Revitalization west (dummy)	2.366	2.516	2.301
	(1.852)	(1.800)	(2.002)
Revitalization Ost (dummy)	−0.224	−0.225	−0.261
	(1.175)	(1.227)	(1.230)
Distance to river (km)	−3.657**	−3.614**	−3.774**
	(0.667)	(0.662)	(1.034)
18–45 year-olds (%)	0.134**	0.129*	0.137**
	(0.049)	(0.053)	(0.052)
CDU (%)	−0.531**	−0.530**	−0.541**
	(0.071)	(0.080)	(0.080)
Government parties (%)	−0.274**	−0.280**	−0.282**
	(0.066)	(0.074)	(0.065)
Turnout (%)	0.169*	0.172*	0.163*
	(0.078)	(0.081)	(0.070)
Appreciation (%)	0.123*	0.183	0.131+
	(0.057)	(0.183)	(0.070)
DuM	−0.871	−0.858	−0.921
	(1.226)	(1.262)	(1.139)
Appreciation (%) × distance to river (km)		−0.035	
		(0.093)	
Constant	21.200*	21.658*	22.325**
	(8.518)	(9.343)	(8.261)
Rho			−0.004
			(0.016)
Observations	87	87	87
(Pseudo) R-squared	0.86	0.86	0.86
Mean VIF	3.89	2.73	3.89
AIC	437.10	438.82	440.93

Notes: Endogenous variable is percentage of yes-votes of eligible voters in all models. Standard errors (in parentheses) are bootstrapped in 500 replication in models (1) and (2).
** $p<0.01$.
* $p<0.05$.

While we find evidence for larger opposition to the project within precincts with relatively higher appreciation rates, this effect can hardly account for the localized support in the referendum, which still significantly increases with proximity to the Mediaspree riverbanks. Note also that in absolute terms, appreciation rates within the Mediaspree development area and the adjoining redevelopment areas were even negative during our observation period. These results are in line with the absence of significant income effects in the voting pattern (see Section 3.1), which also suggests that affordability concerns were not the major driving force for opposition.

We highlight that the Mediaspree riverbank proximity effect is also robust to a range of possible negative externalities, which we account for in a range of separate robustness checks. For instance, the Mediaspree project encompasses a new bridge crossing the river that would possibly generate additional traffic as well as related noise and pollution. As already discussed, a 17,000-seat multifunctional sports arena had been developed within the Mediaspree area. As shown by Ahlfeldt and Maennig (2009) in the case of sports arenas in Berlin-Prenzlauer Berg, which is located only a few kilometers northwards of the study area, these arenas may exhibit both positive as well as negative externalities, in particular on game days. Even the Allianz-Arena in Munich, an architectural landmark stadium designed by Herzog & De Meuron, induced strong localized opposition in a public referendum (Ahlfeldt et al., 2010). The scheduled bridge and the sports arena are located within the core area where construction works will take place and where the related disutility should be expected to be highest. Last, the new development might have been expected to increase overall traffic streams within the study area, which could result in a particularly high opposition along the major traffic arteries. Results for the robustness checks are presented in Table 3 in the appendix. In none of the specifications are the generated spatial variables significant, nor is the distance to the riverbank effect considerably reduced.[13]

This leads us back to the first of the two hypotheses developed at the end of Section 3.1. The localized effects are more likely to be caused by an anticipated disutility associated with the loss of cultural amenities and neighborhood character than by displacement pressures. The disutility decreases with distance as residents' net-consumption benefits decline due to increasing transport costs. In addition, a Tiebout (1956) matching of private cultural goods and unobserved residential preferences may explain the high degree of localized resistance. In order to provide a formal test of this hypothesis, we in the final step of the empirical analysis make use of 360 music nodes in Berlin identified by van Heur (2008), which we georeference based on the provided address data.

These nodes encompass a variety of groups, e.g. music venues, record labels, stores, etc., which are displayed for our study area in Fig. 1. Note that, as discussed, the particular character of the neighborhood is essentially shaped by local music culture. Assuming that the distribution of music nodes serves as a proxy for this specific neighborhood character that activists and voters have been concerned with, we expect an increased opposition in areas with a higher density of cultural activity and the marginal effect of proximity to music nodes to diminish with distance to the Mediaspree. We make use of two indicator variables to capture the music geography, a) the number of music nodes within a walking distance of 1.5 km and b) a music potentiality (*MP*) measure that aggregates music nodes, weighted by distance (*DISTMN*) in km.

$$MN_i = \sum_n \exp(-\tau \times DISTMN_{in}) \tag{6}$$

where τ is a decay parameter determining the weight with which node n enters the potentiality. Similar to Ahlfeldt (2009) and Ahlfeldt and Maennig (2010), we set the decay parameter to a value of 2, which implies an implicit spatial weight function that flats out after approx 2 km in order to reflect walking speed. Results for extended Table 1, column (3) type specifications are presented in Table 3. Only the variables of primary interest are shown to save space since all other coefficients remain qualitatively unchanged. The results relatively clearly confirm our expectations. Opposition increases with the density of cultural activity (columns 1–4) while the marginal effect diminishes with distance to the Mediaspree area as reflected by the negative coefficients on the interactive terms in (3) and (4). Furthermore, the coefficient on the distance to the river is finally rendered insignificant in (3) and (4) and very close to zero in (4), indicating that the spatial heterogeneity in the voting outcome has been accounted for. Notably, the potentiality variable works slightly better than the count variable (and a range of similar variables based on different distance thresholds that were tested), confirming the suitability of potentiality variables to capture complementarities in amenity effects as suggested by Ahlfeldt and Maennig (2010).

Notably, the contribution of the music node potentiality to the explanatory power of the model is considerable. Standardizing Table 3, column (4) coefficients shows that an increase in the potentiality by one standard deviation (S.D.) yields an increase in the dependent variable of 0.53 S.D., more than for any other variable in the model. A regression of the dependent variable on the music node potentiality alone yields and R-squared of 0.54 and thus explains more than one half of the variation in the outcome variable.

A remaining question is whether the distribution of music nodes is causally related to local opposition to the Mediaspree project, or just capturing some preferences that are accidently correlated and have

[13] The only exception is the distance to main roads in column (3). The positive sign, however, is not in line with an increased opposition in proximity to transport arteries. Furthermore, no significant effects are found if the treatment effect is allowed to vary with distance to the riverbank (4).

Table 3
Voting pattern and music nodes.

	(1)	(2)	(3)	(4)	(5)
Referendum	Mediaspree	Mediaspree	Mediaspree	Mediaspree	Tempelhof
Distance to riverbank (km)	−1.947+	−1.729+	−0.67	0.104	1.129*
	(1.139)	(0.904)	(0.790)	(0.758)	(0.523)
Music nodes	0.095*	0.497**	0.166**	0.931**	−0.444
	(0.038)	(0.153)	(0.05)	(0.262)	(0.495)
Music nodes × distance to riverbank			−0.05	−0.294+	−0.185
			(0.034)	(0.159)	(0.195)
Music nodes	Count	Potentiality	Count	Potentiality	Potentiality
Observations	87	87	87	87	87
R-squared	0.86	0.86	0.87	0.88	0.88
Mean VIF	5.12	5.17	5.58	5.77	0.57
AIC	435.606	433.285	432.162	425.140	336.877

Notes:Endogenous variable is percentage of yes-votes of eligible voters in the Mediaspree referendum in (1–4) and the Tempelhof referendum in (5). Music Nodes is number of music nodes within 1.5 km in (1) and (3) and music node potentiality as defined in Eq. (5) in (2) and (4). Robust standard errors are in parentheses.
+ $p<0.1$.
* $p<0.05$.
** $p<0.01$.

not been accounted for by neighborhood controls. To rule out the latter, we conduct a placebo regression on the outcome of the 2008 referendum held in opposition to the closure of Tempelhof Airport, located adjacent to the south western boundary of the study area. Besides being held in the same year, the referenda are comparable in many respects, e.g. in their non-binding nature, the approval by the majority, the relatively low turnout, and a clear position of governing and opposing political parties.[14] We rerun Table 3, column (4) models with a dependent variable generated in exactly the same way as before, i.e. the proportion of eligible voters who supported the referendum. As shown in column (5), music nodes do not impact significantly on the opposition movement reflected in this referendum, indicating that the cultural amenity effect is specific to the Mediaspree referendum and the development project. After all, our results provide strong evidence for the perceived value of (private) cultural amenities, which is receiving increasing attention in the literature (Clark and Kahn, 1988; Hicks and Queen, 2007; Noonan, 2003; Rushton, 2005; Sheppard et al., undated).

4. Conclusion

In this article we investigate the public referendum in opposition to the Mediaspree, a major urban development project in Berlin Friedrichshain–Kreuzberg, against the background of the local evolution of housing prices and the endowment with specific cultural amenities. The referendum was enforced by a movement of neighborhood resistance to perceived pressures of displacement and neighborhood change, including an anticipated threat to the vivid local cultural scene. The project served as a catalyst for fears and anger with regard to a broader process of neighborhood renewal, encompassing two major urban redevelopment programs Stadtumbau West and Stadtumbau Ost that operate adjacent to the Mediaspree area. The resulting conflicts stand exemplary for (re)development strategies brought forth by ambitious authorities, which potentially accelerate regeneration at the cost of displacing the resident population and culture. We add to the literature in a number of respects. First, we provide evidence for a localized resistance to a (re)development strategy which was associated with perceived gentrification, on the basis of the stated preferences of tens of thousands of voters. We find that the opposition was higher in precincts with a large proportion of young adults and residents with a larger interest in policy but a lower affiliation to mainstream parties. Second, we show that there is no compelling evidence for a localized increase in demand for living space in proximity to the renewal area, which should be mirrored in apartment prices. This finding confirms neither a particular success of authorities' revitalization plans nor the stated major concerns of neighborhood activists. Eventually, from an analysis of the impact of precinct level appreciation rates on the voting pattern, our key-findings emerge.

Precincts with a relatively higher appreciation exhibited relatively more opponents engaging in the referendum, conditional on socio-demographic characteristics. This is indicative of a special case of the homevoter hypothesis applying to rental markets where residents oppose an increase in demand for living space. This effect, however, may hardly account for the localized resistance expressed in the referendum, which increases with proximity to the treatment area. In contrast, proxy variables for the specific neighborhood character that captures the local music geography exhibit a significant impact on localized opposition rates and explain the spatial heterogeneity in the voting outcome within the study area.

We conclude that the bulk of the opposition to the urban renewal project is more likely to be caused by an anticipated disutility associated with the loss of a neighborhood character, constituted by specific cultural amenities and related (unobserved) voter preferences, than by perceived displacement pressures. This finding is supported by the absence of significant income effects which we would expect if the affordability of living space was a major concern. We interpret these results as evidence of the perceived value of cultural amenities and intangible neighborhood characteristics, which are often ignored by authorities when developing regeneration strategies. Apparently, the perceived threat of neighborhood change and the displacement of local culture can be as relevant for residents' concerns and resistance as the threat of own displacement, at least in the case of a very specific endowment with cultural goods for which few substitutes are available.

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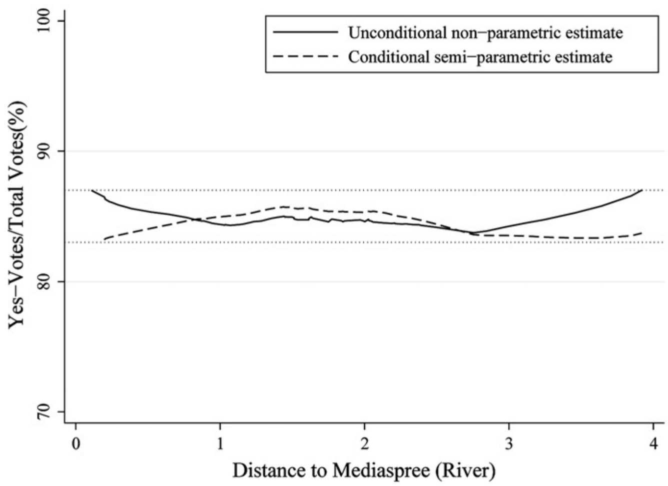
[14] We note that the study area was not considerably affected by aircraft noise. See Nitsch (2009) for details on the referendum.

Appendix A



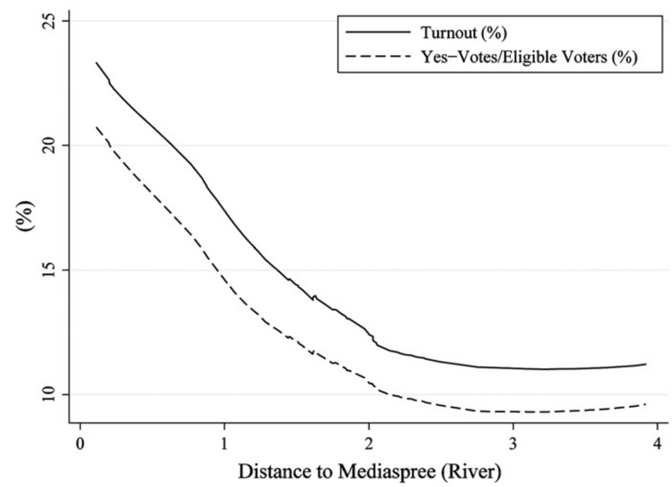


Fig. A1. Turnout and share of yes-votes of eligible voters. Notes: Unconditional gradients are estimated by the use of locally weighted regressions. Semi-parametric estimates use the Lokshin (2006) technique and are conditional on the full set of control variables used in Table 1.

Table A1
Relative trends in apartment prices (full results).

	(1)	(2)	(3)	(4)
Buildings' stories	−0.020	−0.018	−0.019	−0.025*
	(0.012)	(0.012)	(0.013)	(0.013)
Elevator	0.118**	0.121**	0.114**	0.093**
	(0.023)	(0.022)	(0.022)	(0.024)
Below street level	0.051	0.047	0.059	0.007
	(0.049)	(0.049)	(0.049)	(0.053)
Street level	−0.024	−0.023	−0.023	−0.016
	(0.020)	(0.019)	(0.020)	(0.022)
Story	0.014**	0.015**	0.014**	0.012**
	(0.004)	(0.004)	(0.004)	(0.004)
Size (m²)	0.002**	0.002**	0.002*	0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
Size (m²) squared	0	0	0	0
	0	0	0	0
Number of living rooms	0.011*	−0.011*	−0.011*	0.010*
	(0.006)	(0.006)	(0.006)	(0.006)
Vestibule	0.009	0.008	0.006	0.039*
	(0.025)	(0.025)	(0.026)	(0.021)
Attic store room	−0.005	−0.001	−0.006	0.012
	(0.015)	(0.015)	(0.015)	(0.018)
Balcony	0.077**	0.076**	0.075**	0.086**
	(0.016)	(0.016)	(0.017)	(0.014)
Artist studio	0.088*	0.099*	0.104*	0.192*
	(0.049)	(0.044)	(0.050)	(0.074)
Hobby room	0.023	0.029	0.015	−0.017
	(0.183)	(0.175)	(0.181)	(0.240)
Basement room	−0.050*	−0.056**	−0.051*	−0.052*
	(0.021)	(0.021)	(0.022)	(0.023)
Attic storage room	0.076*	0.078*	0.071*	0.029
	(0.037)	(0.037)	(0.039)	(0.045)
Garage	0.194**	0.196**	0.188**	0.171**
	(0.050)	(0.045)	(0.057)	(0.049)
Parking lot	0.066	0.070	0.069	0.064
	(0.063)	(0.064)	(0.061)	(0.067)
Easement at plot	0.048	0.052	0.040	0.058
	(0.071)	(0.072)	(0.068)	(0.058)
Age	−0.003*	−0.003*	−0.002*	−0.005**
	(0.001)	(0.001)	(0.001)	(0.001)
Age squared	0	0	0	0.000*
	0	0	0	0

Table A1 (*continued*)

	(1)	(2)	(3)	(4)
Apartment in bad condition	−0.459**	−0.465**	−0.468**	−0.467**
	(0.069)	(0.069)	(0.070)	(0.067)
Apartment is occupied by renter	0.283**	0.282**	0.277**	0.233**
	(0.025)	(0.024)	(0.025)	(0.026)
Apartment is leased	0.199**	0.198**	0.197**	0.133**
	(0.031)	(0.031)	(0.032)	(0.031)
Social housing	−0.217**	−0.203**	−0.229**	−0.239**
	(0.069)	(0.069)	(0.071)	(0.077)
Tax privileged housing	−0.104	−0.071	−0.087	−0.075
	(0.078)	(0.077)	(0.075)	(0.099)
Authorities nominate tenants	0.055*	0.045*	0.056*	0.066**
	(0.028)	(0.024)	(0.026)	(0.021)
Rent guarantee	0.131**	0.130**	0.137**	0.108**
	(0.021)	(0.023)	(0.024)	(0.020)
Share at joint property	−0.001	−0.001	−0.001	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
Distance to CBD	0.306*	0.107	0.368**	0.277
	(0.137)	(0.195)	(0.137)	(0.187)
Distance to rail station	−0.045	−0.065	−0.019	0.025
	(0.110)	(0.108)	(0.116)	(0.125)
Distance to park	−0.055	−0.107	−0.098	−0.038
	(0.144)	(0.141)	(0.148)	(0.211)
Distance to water	−0.305**	−0.347**	−0.294**	−0.295*
	(0.094)	(0.090)	(0.093)	(0.122)
Distance to school	−0.191*	−0.043	−0.260*	−0.182
	(0.114)	(0.165)	(0.111)	(0.157)
Distance to historic landmark	−0.083	−0.117	−0.131	−0.142
	(0.248)	(0.238)	(0.231)	(0.274)
Constant	7.112**	6.982**	6.728**	8.261**
	(0.051)	(0.113)	(0.266)	(0.214)
Treatment effects	1 km buffer	Distance	Revitalization West & East	
Location effects	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	yes	No
Trend effects	No	No	No	Yes
Period	1997–2008	1997–2008	1997–2008	2000–2008
Observations	9980	9980	9980	7469
R-squared	0.46	0.46	0.46	0.49

Notes: Endogenous variable is log of apartment prices per square meter living area in all models. Robust standard errors (in parentheses) are clustered at precinct level.
* $p<0.05$.
** $p<0.01$.

Table A2
Robustness checks.

	(1)	(2)	(3)	(4)
Distance to river (km)	−3.242	−4.213*	−3.542**	−3.528**
	(1.987)	(1.663)	(0.702)	(0.850)
Distance to new bridge (km)	−0.267			
	(1.429)			
Distance to O2 Arena (km)		0.849		
		(1.293)		
Distance tomain road (km)			0.006*	0.006
			(0.002)	(0.005)
Distance to river × distance to main road				0
				(0.003)
Observations	87	87	87	87
R-squared	0.85	0.85	0.85	0.86

Notes: Endogenous variable is percentage of yes-votes of eligible voters in all models. Baseline model is in Table 1, (3). Only parameters of interested are displayed. Robust standard errors are in parentheses. Mean VIF in all models are <=6.
* $p<0.05$.
** $p<0.01$.

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